Note- File assumed as Plant 2 is- 'Copy of PPD FY25-26

1. ***Overview***

- Plant Type: Rooftop Solar
- Installed Capacity: ~1260 kWp
- Data Period: Apr 2025 – Nov 2025
- Data Source: Remote Monitoring System
- Note: Multiple data blocks present → analysis performed on complete plant dataset (right-side table)

2. ***Key Assumptions***

- GHI (satellite) used as proxy for irradiance
- PR calculated using GHI → used for relative comparison only
- Underperformance threshold → Generation < 2000 kWh
- Loss values are estimated, not exact
- Inverter availability assumed based on data presence

3. ***Key Performance Summary***

Month	CUF	PR	Active Inv	Underperf Days
Apr	9.6%	0.41	7	3
May	9.7%	0.49	7	4
Jun	11.1%	0.72	10	9
Jul	9.2%	0.80	10	11
Aug	9.8%	0.78	10	2
Sep	9.2%	0.78	10	8
Oct	12.8%	0.74	10	1
Nov	12.8%	0.72	10	0

4. *Major Findings*

Best Performance is observed in-

- **October & November**
    - CUF ~12.8%
    - Stable PR (~0.72–0.74)
    - Minimal underperformance

Worst Performance

- **April & May**
    - CUF ~9.6%
    - PR very low (~0.41–0.49)
    - Only 70% capacity active

5. *Capacity Utilization Analysis*

April–May (Critical Issue is observed)

- Active Inverters → **7/10**
- ~30% capacity inactive
- Significant generation loss

Impact

- April generation → ~87 MWh
- Expected (full capacity) → ~125 MWh

Estimated loss: ~37 MWh due to inactive inverters

Conclusion- Low CUF and PR in early months are driven by capacity underutilization, not system inefficiency.

June (Transition Phase)

- All inverters active (10/10)
- Minor 2-day inverter issue
- Performance improves significantly

July–November (Stable Phase)

- Full capacity utilization
- Consistent inverter availability
- No structural issues

6. *Underperformance Analysis*

Summary showing total underperformance inverters segregated into system issues and weather issues

Month	**Total**	**System**	**Weather**
Apr	3	2	1
May	4	1	3
Jun	9	2	7
Jul	11	0	11
Aug	2	0	2
Sep	8	0	8
Oct	1	0	1
Nov	0	0	0

Key Insights

- Early months - mixed (system + weather)
- Later months - mostly weather-driven
- System-related issues are minimal

7. *PR Analysis*

- Apr–May → very low PR (0.41–0.49)
- Jun onwards → normal PR (0.72–0.80)

Low PR in early months is majorly due to inactive inverters.

Benchmark Comparison

- First-Year PR (from master) - ~75%
- Current PR (post June) - ~72–80%

Conclusion- No significant degradation observed

8. *CUF Trend Analysis*

Observations

- Apr–May - low CUF (~9.6%)
- Jun–Sep - moderate (~9–11%)
- Oct–Nov - highest (~12.8%)

So, CUF improves after full commissioning and peaks in high irradiance months

9. *Seasonal Impact*

- July–September - high underperformance days
- Driven by weather (monsoon)
- PR remains stable so the system is healthy

So, performance variation is seasonal, not operational

10. *Loss Breakdown for showing major Loss Contributors*

1. Capacity Loss (PRIMARY)

- April - ~37 MWh
- May - similar impact

2. Weather Loss

- July– September
- High number of low-generation days

3. System Loss

- Minimal (few days only)

Final Loss Insight- Capacity underutilization is the dominant loss driver

11. *Final Key Takeaways*

- Early months impacted by **missing inverters (~30% capacity)**
- Full commissioning achieved from June onwards
- Weather dominates losses in monsoon months
- Minimal persistent system faults
- Strong performance in Oct–Nov
- No long-term degradation observed

12. *Recommendations for improvement*

- Ensure complete commissioning before performance evaluation
- Monitor inverter-level availability in early stages
- Implement availability tracking KPI
- Improve forecasting during monsoon months